Principles of Programming Languages
Semantics. An evaluator for IMP.

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Outline

Semantics: introduction

An evaluator for IMP

Semantics

- Semantics is concerned with the meaning of language constructs
- Semantics must be unambiguous
- Semantics must be flexible

Semantic - informal

Informal semantics (examples): natural language

Rationale for the ANSI C Programming Language:

- "Trust the programmer"
- "Don't prevent the programmer from doing what needs to be done"
- "Keep the language small and simple"
- "Provide only one way to do an operation"
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Formal Semantics

Some (formal) semantics styles:

- operational
- denotational
- axiomatic

We will focus more on operational semantics styles: Small-step SOS, Big-Step SOS

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A simple imperative language

Program:

```
n ::= 10;;
i ::= 1;;
sum ::= 0;;
while (i «= n) do
     sum ::= sum +′ i;;
     i ::= i +′ 1
end
```

Features:

- variables
- arithmetic expressions
- boolean expressions
- assignment statements
- loop statements
- decisional statements

How can we define the semantics of this language?

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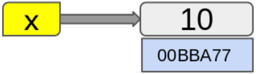
- variable = *storage location* at *address* + *symbolic name*
- variables can be *accessed* or *modified*
- depending on the language, they may only be able to store a specified datatype (e.g., integer, string, etc.)
- *Scope*: global, local
- Lifetime (*extent*)
- $var\_name \mapsto value$
- Environment + memory



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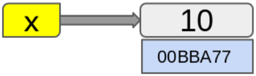

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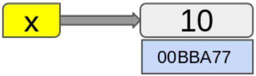

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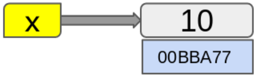

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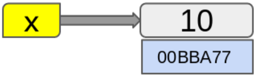

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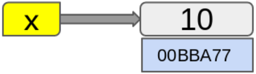

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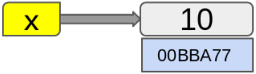

Environment

- Environment: variables mapped to values

```
Definition Env := string -> nat.
Definition env (string : string) :=
if (string_dec string "n")
     then 10
     else 0.
```

- The environment is just a function

```
Compute (env n).
= 10
: nat
Compute (env i).
= 0
: nat
Check env.
env : string -> nat
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Environment update

- Update = a new function s.t. the value for `x` is updated to `v`:

```
Definition update (x : string)
                  (v : nat)
                  (env : Env) : Env :=
    fun y => if (string_dec x y)
             then v
             else env y.
```

Arithmetic expressions

- Syntax:

BNF:

$$\begin{array}{lcl} \text{AExp} & ::= & \mathbb{N} \\ & \mid & \text{AExp `+' AExp} \\ & \mid & \text{AExp `*' AExp} \end{array}$$

```
Inductive AExp :=
| anum : nat -> AExp
| aplus : AExp -> AExp -> AExp
| amul : AExp -> AExp -> AExp.
```

- Semantics:

```
Fixpoint aeval (a : AExp) : nat :=
  match a with
  | anum v => v
  | aplus a1 a2 => (aeval a1) + (aeval a2)
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Arithmetic expressions with *variables*

- Syntax:

BNF:

AExp	::=	string
	|	$\mathbb{N}$
	|	AExp '+' AExp
	|	AExp '*' AExp

```
Inductive AExp :=
| var : string -> AExp
| anum : nat -> AExp
| aplus : AExp -> AExp -> AExp
| amul : AExp -> AExp -> AExp.
```

- Semantics:

```
Fixpoint aeval (a : AExp) : nat :=
  match a with
  | var v => ???
  | anum v => v
  | aplus a1 a2 => (aeval a1) + (aeval a2)
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```

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Evaluate expression with variables

- Semantics:

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Fixpoint aeval (a : AExp) (env : Env) : nat :=
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  end.
```

- Notations and testing:

```
Coercion var : string >-> AExp.
Coercion anum : nat >-> AExp.
Notation "A +' B" := (aplus A B) (at level 50).
Notation "A *' B" := (amul A B) (at level 40).
Compute aeval (2 +' 3 *' 4) env.
= 14
: nat
Compute env n.
= 10
: nat
Compute aeval (2 +' 3 *' n) env.
= 32
: nat
```

Boolean expressions

- Syntax:

```
Inductive BExp :=
| btrue : BExp
| bfalse : BExp
| blessthan : AExp -> AExp -> BExp
| band : BExp -> BExp -> BExp
| bnot : BExp -> BExp.
```

- Semantics:

```
Fixpoint beval (b : BExp) (env : Env) : bool :=
  match b with
  | btrue => true
  | bfalse => false
  | blessthan a1 a2 => Nat.leb (aeval a1 env) (aeval a2 env)
  | band b1 b2 => andb (beval b1 env) (beval b2 env)
  | bnot b' => negb (beval b' env)
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  end.
```

Assignments

- Syntax:

```
Inductive Stmt :=
| assignment : string -> AExp -> Stmt.
Notation "A ::= B" := (assignment A B) (at level 54).
Check n ::= 100 .
n ::= 100
: Stmt
```

- Assignments modify the environment
- Semantics:

```
Fixpoint eval(s : Stmt) (env : Env) : Env :=
match s with
| assignment x a => update x (aeval a env) env
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Programs: sequence of statements

- Syntax:

```
Inductive Stmt :=
| seq s1 s2 => eval s2 (eval s1 env)
Notation "S S'" := (assignment S S') (at level 54).
Check n ::= 100 ;; i ::= 7 .
n ::= 100 ;;x i ::= 7
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Loops

- Syntax:

```
Inductive Stmt :=
| while : BExp -> Stmt -> Stmt.
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        Fixpoint eval (s : Stmt) (env : Env) : Env :=
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  end.
```

- ERROR:

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Error: Cannot guess decreasing argument of fix.
```

- (Partial) solution: add a *fuel* parameter
- The evaluation can also be defined as a *relation* (future lecture)

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Evaluation as a relation

- Assignment: $\dfrac{\texttt{aeval a E} = \texttt{v}}{(\texttt{eval}\ (\texttt{x} ::= \texttt{a})\ \texttt{E}\ (\texttt{x} \mapsto \texttt{v}\ ;\ \texttt{E}))}$
- Sequence: $\dfrac{\texttt{eval}\ \texttt{s}_1\ \texttt{E}_1\ \texttt{E}' \quad \texttt{eval}\ \texttt{s}_2\ \texttt{E}'\ \texttt{E}_2}{(\texttt{eval}\ (\texttt{seq}\ \texttt{s}_1\ \texttt{s}_2)\ \texttt{E}_1\ \texttt{E}_2)}$
- Loop (true case):

$$\frac{\texttt{beval b}\ \texttt{E}_1 = \texttt{true} \quad \texttt{eval s}\ \texttt{E}_1\ \texttt{E}' \quad \texttt{eval}\ (\texttt{while b s})\ \texttt{E}'\ \texttt{E}_2}{(\texttt{eval}\ (\texttt{while b s})\ \texttt{E}_1\ \texttt{E}_2)}$$

- Loop (false case): $\dfrac{\texttt{beval b E} = \texttt{false}}{(\texttt{eval}\ (\texttt{while b s})\ \texttt{E E})}$

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- Sequence: $\dfrac{\texttt{eval}\ \texttt{s}_1\ \texttt{E}_1\ \texttt{E}' \quad \texttt{eval}\ \texttt{s}_2\ \texttt{E}'\ \texttt{E}_2}{(\texttt{eval}\ (\texttt{seq}\ \texttt{s}_1\ \texttt{s}_2)\ \texttt{E}_1\ \texttt{E}_2)}$
- Loop (true case):

$$\frac{\texttt{beval b}\ \texttt{E}_1 = \texttt{true} \quad \texttt{eval s}\ \texttt{E}_1\ \texttt{E}' \quad \texttt{eval}\ (\texttt{while b s})\ \texttt{E}'\ \texttt{E}_2}{(\texttt{eval}\ (\texttt{while b s})\ \texttt{E}_1\ \texttt{E}_2)}$$

- Loop (false case): $\dfrac{\texttt{beval b E} = \texttt{false}}{(\texttt{eval}\ (\texttt{while b s})\ \texttt{E E})}$

Evaluation as a relation

- Assignment: $\dfrac{\texttt{aeval a E} = \texttt{v}}{(\texttt{eval}\ (\texttt{x} ::= \texttt{a})\ \texttt{E}\ (\texttt{x} \mapsto \texttt{v}\ ;\ \texttt{E}))}$

- Sequence: $\dfrac{\texttt{eval}\ \texttt{s}_1\ \texttt{E}_1\ \texttt{E}' \quad \texttt{eval}\ \texttt{s}_2\ \texttt{E}'\ \texttt{E}_2}{(\texttt{eval}\ (\texttt{seq}\ \texttt{s}_1\ \texttt{s}_2)\ \texttt{E}_1\ \texttt{E}_2)}$

- Loop (true case):

$$\frac{\texttt{beval b}\ \texttt{E}_1 = \texttt{true} \quad \texttt{eval s}\ \texttt{E}_1\ \texttt{E}' \quad \texttt{eval}\ (\texttt{while b s})\ \texttt{E}'\ \texttt{E}_2}{(\texttt{eval}\ (\texttt{while b s})\ \texttt{E}_1\ \texttt{E}_2)}$$

- Loop (false case): $\dfrac{\texttt{beval b E} = \texttt{false}}{(\texttt{eval}\ (\texttt{while b s})\ \texttt{E E})}$

Bibliography

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